

Analysis General Exam August 2017

Instructions. 4 hours. Closed book examination. As the exam is a bit long you do not have to do everything in order to succeed. In order to pass, you need to complete a significant portion of each of the two sections: complex analysis (Questions 1 through 4) and real analysis (Questions 5 through 7). Fully completing one section while leaving the other untouched will not result in a passing grade so please plan the allocation of your time and effort accordingly.

Problem 1

Let f be an entire function. Which of the following conditions imply that f is constant? Give proofs or counterexamples.

(a) $\operatorname{Re} z \geq 0 \Rightarrow |f(z)| \leq 1.$

(b) $|z| \geq 1 \Rightarrow \left| \frac{f(z)^2}{z} \right| \leq \pi.$

Problem 2

Evaluate

$$\int_0^\infty \frac{1}{x^n + 1} dx$$

where $n \geq 2$ is an integer. Your final answer should not involve any reference to complex numbers.

Problem 3

(a) Let f be an analytic function. Show that the level curves $\operatorname{Re} f(z) = k_1, \operatorname{Im} f(z) = k_2$ are perpendicular at any point where $f'(z) \neq 0$.

(b) Let $p_1, p_2, \dots, p_n$ be points on a circle of radius 1. Show that there is a point on the circle such that the product of its distances to the p_j is 1. (Suggestion: apply the maximum modulus principle to an appropriate complex polynomial.)

Problem 4

Let z_1, z_2 be unequal numbers in the open unit disk $\mathbb{D}$, and let f be an analytic selfmap of $\mathbb{D}$, not necessarily 1-1. Follow the steps below to prove the following version of Pick's Lemma:

$$\left| \frac{f(z_1) - f(z_2)}{1 - \overline{f(z_1)}f(z_2)} \right| \leq \left| \frac{z_1 - z_2}{1 - \overline{z_1}z_2} \right|$$

- (a) Write down an explicit conformal self-map of the disk that sends $f(z_2)$ to 0. You do not need to prove anything about your map.
- (b) Find conformal self-maps of the disk φ, ψ such that $\varphi \circ f \circ \psi^{-1}$ is an analytic self-map of the disk sending 0 to 0, and apply the Schwarz lemma appropriately to $\varphi \circ f \circ \psi^{-1}$.

Problem 5

- (a) Recall that continuous functions in $L^2(\mathbb{R})$ are dense in $L^2(\mathbb{R})$. Show that compactly supported continuous functions are dense in $L^2(\mathbb{R})$.
- (b) For positive a and b let $f_{a,b} = a\chi_{[0,b]}$. Produce a sequence (a_n, b_n) such that $\int_{\mathbb{R}} f_{a_n, b_n}(x)dx = 1$ for all n while $\|f_{a_n, b_n}\|_{L^2} \rightarrow 0$ when $n \rightarrow \infty$.
- (c) Show that compactly supported continuous functions f such that $\int_{\mathbb{R}} f(x)dx = 0$ are dense in $L^2(\mathbb{R})$.
- (d) Are continuous functions f such that $\int_{[0,1]} f(x)dx = 0$ dense in $L^2([0,1])$? Justify your answer.

Problem 6

- (a) Let $h_{0,0}(x) = \chi_{[0, \frac{1}{2}]}(x) - \chi_{[\frac{1}{2}, 1]}(x)$. More generally, for $n \geq 0$ and $0 \leq k \leq 2^n - 1$, we define $h_{n,k}(x) = 2^{\frac{n}{2}} h_{0,0}(2^n x - k)$. For $n \geq 0$ and $(x, y) \in [0, 1]^2$ we let

$$L_n(x, y) = \sum_{k=0}^{2^n-1} h_{n,k}(x)h_{n,k}(y)$$

and

$$K_n(x, y) = \chi_{[0,1]}(x)\chi_{[0,1]}(y) + \sum_{m=0}^n L_m(x, y)$$

Plot L_0, L_1, L_2 and K_0, K_1, K_2 . (Do not use 3d plots but simply subdivide the square into regions and in each region indicate the value of the function. Do six different plot sketches, one for each function).

- (b) Show that for $n \geq 0$ and $(x, y) \in [0, 1]^2$, one has $K_n(x, y) = 2^{n+1}$ if x, y both belong to $[\frac{k}{2^{n+1}}, \frac{k+1}{2^{n+1}})$ for some $k, 0 \leq k \leq 2^{n+1} - 1$, and $K_n(x, y) = 0$ otherwise.
- (c) Let f be continuous on $[0, 1]$ and, for $n \geq 0$, define

$$g_n(x) = \int_0^1 K_n(x, y) f(y) dy$$

Show that $\lim_{n \rightarrow \infty} g_n = f$ in $L^p([0, 1])$ for all $p \in [1, \infty)$.

(d) Let V be the linear span of the collection of functions $h_{n,k}$, with $n \geq 0, 0 \leq k \leq 2^n - 1$, together with the constant function equal to 1. Prove that V is dense in $L^p([0, 1])$ for all $p \in [1, \infty)$.

(e) In the particular case $p = 2$, give a geometric interpretation for the map which produces g_n out of f .

Problem 7

(a) In this problem the formula

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^n e^{-\frac{x^2}{2}} dx = 1$$

for $n = 0$ and $n = 2$ can be used without justification. For $(t, x) \in (0, \infty) \times \mathbb{R}$ we let

$$P(t, x) = \frac{1}{2\sqrt{\pi t}} e^{-\frac{x^2}{4t}}$$

Show that this function is infinitely differentiable and that

$$\frac{\partial^{m+n} P}{\partial t^m \partial x^n}(t, x) = \frac{Q_{m,n}(t, x)}{2\sqrt{\pi t}^{2m+n+\frac{1}{2}}} e^{-\frac{x^2}{4t}}$$

for some two-variable polynomials $Q_{m,n}(t, x)$ satisfying the recursion

$$\begin{aligned} Q_{m+1,n} &= t^2 \frac{\partial Q_{m,n}}{\partial t} + \left(\frac{x^2}{4} - (2m + n + \frac{1}{2})t \right) Q_{m,n} \\ Q_{m,n+1} &= t \frac{\partial Q_{m,n}}{\partial x} - \frac{x}{2} Q_{m,n} \end{aligned}$$

(b) Let f be a compactly supported twice continuously differentiable function on $\mathbb{R}$. Show that

$$\psi(t, x) = \int_{\mathbb{R}} P(t, x - y) f(y) dy$$

is well defined and infinitely differentiable on $(0, \infty) \times \mathbb{R}$.

(c) By computing $Q_{0,0}, Q_{1,0}, Q_{0,1}, Q_{0,2}$ show that ψ satisfies the heat equation, i.e.,

$$\frac{\partial \psi}{\partial t} = \frac{\partial^2 \psi}{\partial x^2}$$

on $(0, \infty) \times \mathbb{R}$.

(d) Recall that a twice continuously differentiable function f satisfies the Taylor formula with integral remainder

$$f(y) = f(x) + (y - x)f'(x) + (y - x)^2 \int_0^1 (1 - s)f''(x + s(y - x))ds$$

Extend ψ to $[0, \infty) \times \mathbb{R}$ by letting $\psi(0, x) = f(x)$ for all x . Show that the heat equation continues to hold for this extension with $\frac{\partial}{\partial t}$ derivatives at $t = 0$ understood as right-derivatives. Hint: use the previous Taylor formula and a suitable change of variable in order to establish the identity

$$\frac{\psi(t, x) - \psi(0, x)}{t} = \frac{1}{2\sqrt{\pi}} \int_{[0,1]} \int_{\mathbb{R}} z^2 e^{-\frac{z^2}{4}} (1 - s)f''(x + s\sqrt{t}z)dzds$$

for $t > 0$.